

ALEX ABABU

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• Authorized to work in the US • No Visa Sponsorship Required • Open to Relocate

WORK EXPERIENCE

AI Acceleration Analyst, Arizona State University Aug 2025 – Present

- Partnered with university stakeholders to identify workflow bottlenecks, gather requirements, and deliver AI-powered software solutions that improved university operations
- Built a Slack-based AI assistant on AWS (Lambda, API Gateway, SQS, DynamoDB) that automated student and faculty support requests across multiple university departments
- Developing a multimodal AI evaluation platform using Python, React, and Gemini 3.5 Flash to assess videos, PDFs, DOCX, and presentations against custom rubrics, leveraging LLM-as-a-judge workflows to extract evidence, generate rubric-based scores, and automate batch evaluations that saves staff and faculty hours of manual review
- Wrote advanced SQL queries in Amazon Redshift to generate reports for university leadership and built Python automation to detect data inconsistencies between databases before they affected reporting

Software Engineer, Freelance: Jan 2024 – Aug 2025

- Built a license plate recognition module for an automated vehicle gate system using OpenCV for image preprocessing and a CNN-based OCR model for character recognition, handling check-in/check-out and reporting end to end against client specs
- Built a real estate listing app in Flutter, implementing a pixel-perfect UI from Figma along with authentication and property search, and wiring up GraphQL and Firebase for backend services

Solutions Architect, LonAdd Consultancy Feb 2023 – Jan 2024

- Translated business requirements into user stories, API specifications, UML diagrams, and UI prototypes (Figma, Balsamiq), enabling engineering teams to deliver features for a large-scale mobile money platform
- Coordinated end-to-end user acceptance testing (UAT), defect tracking, and release readiness, helping deliver production releases on schedule
- Collaborated with business stakeholders, banking partners, and engineering teams to define, prioritize, and document product requirements throughout the software development lifecycle

Software Developer, Arifpay Financial Technologies Dec 2021 – Feb 2023

- Designed a configurable payment processing platform that supported the messaging standards of multiple banks, replacing six separate applications with a single solution and significantly reducing maintenance and deployment effort
- Built a common software layer that allowed the payment system to work across multiple POS device vendors, eliminating duplicate code and simplifying support for new hardware
- Modernized legacy Java services by redesigning the application architecture, reducing code complexity by nearly 90% and improving payment authorization time from 60 seconds to 2 seconds
- Developed Spring Boot APIs for a payment gateway used by 20+ financial institutions, while leading a team of five engineers in Agile development and mentoring junior developers

SKILLS

- **Languages:** Python, Java, TypeScript, SQL
- **Cloud & AWS:** AWS Lambda, API Gateway, SQS, S3, EC2, DynamoDB, Redshift, Bedrock, AWS SAM
- **Backend:** Spring Boot, FastAPI, REST APIs, Serverless & Event-Driven Architecture, Docker
- **Databases & Data:** SQL, PostgreSQL, DynamoDB, ChromaDB, OpenSearch, Neo4j, pgvector, PySpark
- **AI & LLM:** LangChain, CrewAI, RAG, GraphRAG, MCP, Hugging Face, Prompt Engineering, LLM APIs, Function Calling, Structured Outputs (Pydantic), Embeddings, Semantic Search
- **Software Engineering:** Git/GitHub, CI/CD, Unit & Integration Testing, Linux, Agile/Scrum, Code Reviews
- **Security & Tools:** Microsoft Presidio, PII Masking, RBAC, Claude Code, Jira

EDUCATION

Master of Science in Artificial Intelligence in Business, Arizona State University, Tempe, AZ May 2026

Relevant Coursework: AWS, NLP, Computer Vision, Machine Learning, AI Agents

4.0 GPA

Bachelor of Science in Computer Science, Unity University Nov 2021

Relevant Coursework: Data Structures and Algorithms, Object Oriented Programming, Database Management

3.82 GPA

PROJECTS

Credit Risk Prediction (Python, PySpark, LightGBM)

- Built a distributed data pipeline in PySpark to clean and transform financial transaction data for machine learning.
- Trained and evaluated a LightGBM model to predict customer credit risk

Privacy-Aware Semantic Retrieval (Python, ChromaDB, Streamlit, Microsoft Presidio, Pydantic)

- Built a data pipeline to ingest, anonymize, and index large conversation datasets for semantic search using vector embeddings
- Implemented privacy-preserving search with PII masking, ranked retrieval, and configurable similarity thresholds

Geospatial Fleet Management Platform: (Node.js, TypeScript, React, REST APIs, AWS EC2)

- Built a full-stack fleet management platform with REST APIs, vehicle dispatch services, and an administrative dashboard
- Integrated Google Maps services and deployed the application on AWS EC2 running Linux